

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)

Data structure

Practical – 4

Queue Application:

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Queue Application:

a. Write a program to: a. Implement a queue using an array.

Input

```
from array import array
queue = array("i")
queue.append(10)
queue.append(20)
queue.append(30)
print("Queue:", queue)
queue.pop(0)
print("After Dequeue:", queue)
print("S012 Priyanshi Gupta")
```

Output

```
Queue: array('i', [10, 20, 30])
After Dequeue: array('i', [20, 30])
S012 Priyanshi Gupta
```

b. Simulate a simple queuing system (e.g., customer service queue).

Input

```
import time
que = [("Priya", 2), ("Aashi", 3), ("Shraddha", 5), ("Kirti", 7)]
print("-----Booking ticket-----")
print(que)
while que:
    job, timer = que.pop(0)
    print(f"Running: {job} {timer} sec")
    time.sleep(timer)
print(que)
print("S012 Priyanshi Gupta")
```

Output

```
-----Booking ticket-----  
[('Priya', 2), ('Aashi', 3), ('Shraddha', 5), ('Kirti', 7)]  
Running: Priya2sec  
Running: Aashi3sec  
Running: Shraddha5sec  
Running: Kirti7sec  
[]  
S012 Priyanshi Gupta
```